Exercise Sheet

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

The core uses finite-dimensional inner-product spaces over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ in the physics convention (conjugate-linear in the first slot). Tags mark the flavour; a ★ marks a harder one.

Core tutorial route

Work **8.2, 8.3, 8.5, 8.17, 8.8, 8.9, and 8.19** in that order (about 80–100 minutes before discussion). This seven-problem route covers coordinates and Parseval, real and complex Gram–Schmidt, an orthogonal complement/dimension check, best approximation, least-squares transfer, and a genuine finite Riesz explanation with the complex bra–ket convention.

Additional practice: 8.1, 8.4, 8.6–8.7, 8.10–8.14, 8.16, 8.18, 8.20, 8.24. **Bridge:** 8.15, 8.21–8.22 (integral/Frobenius models and a locally defined adjoint preview). **Extension:** 8.23 (codimension-one forward link). Extension tasks do not count toward core progress.

Orthonormal bases and Gram–Schmidt

Exercise 8.1 (computational) Apply Gram–Schmidt to $v_1 = (1,1,1)$, $v_2 = (1,1,0)$, $v_3 = (1,0,0)$ in $\mathbb{R}^3$ with the dot product.

Exercise 8.2 (computational) Expand $v = (2,3) \in \mathbb{R}^2$ in the orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1,1)$, $e_2 = \frac{1}{\sqrt{2}}(1,-1)$, and verify Parseval's relation.

Exercise 8.3 (computational) Apply Gram–Schmidt to $v_1 = (3,4)$, $v_2 = (1,0)$ in $\mathbb{R}^2$ with the dot product.

Exercise 8.4 (computational) Expand $v = (3,0,0) \in \mathbb{R}^3$ in the orthonormal basis $e_1 = \frac{1}{3}(1,2,2)$, $e_2 = \frac{1}{3}(2,1,-2)$, $e_3 = \frac{1}{3}(2,-2,1)$, and check Parseval's relation.

Exercise 8.5 (computational) Orthonormalize $v_1 = (1,i)$, $v_2 = (0,1)$ in $\mathbb{C}^2$ (physics convention, conjugate-linear in the first slot).

Exercise 8.6 (proof) Prove that an orthonormal list $e_1, \dots, e_m$ is linearly independent directly from the definition.

Exercise 8.7 (proof) Let $e_1, \dots, e_n$ be an orthonormal basis of V. Prove that for all $u, v \in V$,

$$\langle u, v\rangle = \sum_{k=1}^{n} \overline{\langle e_k, u\rangle}\,\langle e_k, v\rangle,$$

so the coordinate map $v \mapsto (\langle e_1, v\rangle, \dots, \langle e_n, v\rangle)$ preserves inner products.

Projections and best approximation

Exercise 8.8 (computational) Find the point of the line $U = \operatorname{span}\big((1,2,2)\big) \subset \mathbb{R}^3$ closest to $b = (3,0,0)$.

Exercise 8.9 (computational) Fit a line $y = c_0 + c_1 x$ to the data $(1, 1)$, $(2, 2)$, $(3, 2)$ by least squares.

Exercise 8.10 (computational) Find the orthogonal projection of $b = (2, 4) \in \mathbb{R}^2$ onto the line $U = \operatorname{span}\big((1, 1)\big)$, and the distance from b to U.

Exercise 8.11 (computational) Project $b = (3, 0, 0) \in \mathbb{R}^3$ onto the plane U with orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Exercise 8.12 (computational) Solve the inconsistent system $Ax = b$ in the least-squares sense, where $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $b = (1, 1, 0)$.

Exercise 8.13 (proof) Suppose $P = P_U$ is an orthogonal projection. Prove $P^2 = P$ and that $v - Pv \perp Pv$ for every v.

Exercise 8.14 (proof) Let P_U be the orthogonal projection onto a subspace U with orthonormal basis $e_1, \ldots, e_m$. Prove that $\langle P_U v, w\rangle = \langle v, P_U w\rangle$ for all $v, w \in V$.

Exercise 8.15 ($\star$) On $\mathcal{P}_2$ with $\langle p, q\rangle = \int_{-1}^{1} \overline{p}\, q\, dx$, find the polynomial of degree ≤ 1 closest to $f(x) = x^2$, and verify that the error is orthogonal to $\mathcal{P}_1$.

Riesz and orthogonal complements

Exercise 8.16 (computational) Find the vector $w \in \mathbb{C}^2$ representing the functional $\varphi(v) = 2v_1 - iv_2$ in the form $\varphi(v) = \langle w, v\rangle$.

Exercise 8.17 (computational) For $U = \operatorname{span}\big((1, 0, 1)\big) \subset \mathbb{R}^3$, describe $U^\perp$ and verify $\dim U + \dim U^\perp = 3$.

Exercise 8.18 (computational) Find $w \in \mathbb{R}^3$ representing the functional $\varphi(v) = 2v_1 - v_2 + 3v_3$ in the form $\varphi(v) = \langle w, v\rangle$.

Exercise 8.19 (proof) State and prove the finite-dimensional Riesz representation theorem for a linear functional $\varphi \colon V \to \mathbb{F}$, including its hypotheses, existence, and uniqueness. Then explain why $w \mapsto \langle w,\ cdot\ \rangle$ is conjugate-linear over $\mathbb{C}$ but linear over $\mathbb{R}$.

Exercise 8.20 (computational) For $U = \operatorname{span}\big((1, 0, 1, 0), (0, 1, 0, 1)\big) \subset \mathbb{R}^4$, give a basis of $U^\perp$ and check $\dim U + \dim U^\perp = 4$.

Exercise 8.21 (computational) On $\mathcal{M}_2(\mathbb{R})$ with the Frobenius product $\langle A, B\rangle = \operatorname{tr}(A^\dagger B)$, find the matrix w representing $\varphi(A) = 2A_{11} - A_{12} + A_{22}$ as $\varphi(A) = \langle w, A\rangle$.

Exercise 8.22 ($\star$) **Forward bridge.** For a matrix/operator T in an ONB, define its adjoint $T^\dagger$ by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ (equivalently, conjugate transpose of the matrix). Prove that $\operatorname{tr}(T^\dagger T) = \sum_{j,k} |\langle e_j, Te_k\rangle|^2 \geq 0$, with equality iff $T = 0$.

Exercise 8.23 ($\star$) Suppose V is finite-dimensional and φ is a linear functional on V represented by w via $\varphi(v) = \langle w, v\rangle$. Prove that $\operatorname{null} \varphi = (\operatorname{span}(w))^\perp$ and, if $w \neq 0$, that $\dim \operatorname{null} \varphi = \dim V - 1$.

Exercise 8.24 (proof) Let U be a subspace of an inner product space V. Prove directly from the definitions that $U \subseteq (U^\perp)^\perp$.